COURSE OUTLINE

Agentic Attack Surface Management

Discover, reconcile, contextualize and assign exposed assets

- 3 days (8 hours per day)
- Intermediate
- Instructor-led, hands-on labs
- Capstone project

About this course

Who should attend

External attack surface teams, security operations, cloud security, IT asset management, MSSPs.

Prerequisites

- DNS, HTTP and TLS fundamentals
- Asset inventory concepts
- Python
- Basic cloud and network knowledge

What you will be able to do

- Build a normalized asset inventory from approved sources with explicit scope enforcement
- Reconcile domains, certificates, cloud resources, services and ownership records
- Detect inventory changes and potential stale or orphaned assets
- Correlate assets with vulnerability and exposure findings, separating observed facts from inferred exposure
- Generate evidence-backed ownership and remediation tasks while blocking out-of-scope discovery

Tools and technologies

- Claude
- Python
- FastAPI
- PostgreSQL
- Certificate transparency
- DNS (fixtures)
- Cloud inventory (fixtures)
- Azure inventory (fixtures)
- Jira (mock)
- pytest
- Docker Compose

Table of contents

DAY 1 3 modules

MODULE 1

Authorization, scope policy and the lab

- Explicit allowlist of lab-owned domains and synthetic assets, exclusions, rate limits, audit trail
- Arbitrary target entry is disabled in the course environment
- Why weak scope is the biggest risk in attack surface tooling

HANDS-ON LAB
Implement the scope policy and prove that out-of-scope targets are blocked and logged.

DELIVERABLE
Scope policy with audit.

MODULE 2

Source adapters

- Fixture-based adapters for DNS records, certificate metadata, cloud inventory, CMDB entries and scanner findings
- Adapter contract, timeouts, retries and least privilege

HANDS-ON LAB
Build and test the five adapters against fixtures.

DELIVERABLE
Source adapter layer.

MODULE 3

Asset normalization and canonical IDs

- Canonical asset IDs; source, first and last seen, confidence, environment and relationships
- Provenance on every record

HANDS-ON LAB
Normalize all sources into the inventory store.

DELIVERABLE
Normalized asset inventory.

DAY 2 3 modules

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MODULE 4

Entity resolution

- Deterministic signals: exact hostname, certificate SAN, cloud resource ID, verified ownership
- Uncertain matches go to a review queue, never silently merged
- Precision-oriented tests

HANDS-ON LAB
Implement entity resolution with a review queue and measure link precision.

DELIVERABLE
Entity resolution with review queue.

MODULE 5

Change detection

- Snapshot comparison: new, changed, missing, stale and potentially orphaned assets
- Synthetic fixtures with known expected outcomes

HANDS-ON LAB
Build snapshot comparison and validate it against the fixture set.

DELIVERABLE
Change detection engine.

MODULE 6

Exposure correlation with an agent

- Joining assets to scanner findings, service metadata and business criticality
- The agent summarizes exposure but must distinguish observed facts from inferred exposure
- Unsupported exposure claims are measured

HANDS-ON LAB
Correlate exposure and review the agent's output for unsupported claims.

DELIVERABLE
Exposure correlation.

DAY 3 3 modules

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MODULE 7

Ownership resolution

- Synthetic CMDB records, tags and repository metadata to suggest a responsible team
- Evidence attached to every suggestion; manual correction supported

HANDS-ON LAB
Resolve ownership for the inventory and correct three wrong suggestions.

DELIVERABLE
Ownership resolver.

MODULE 8

Prioritization, ticketing and inventory quality metrics

- Transparent ranking criteria and review tickets through a mock connector
- Entity-link precision, unowned asset rate, stale-asset classification, change-detection accuracy
- Asset graph, reports and audit

HANDS-ON LAB
Rank, raise review tickets and publish the quality metrics.

DELIVERABLE
Metrics report and asset graph.

MODULE 9

Capstone: scoped attack surface platform

- Approved sources to owned, prioritized exposures in one run
- Acceptance criteria, limitations, demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end attack surface management lab.

Capstone

Capstone project

A platform that consolidates approved asset sources, detects changes and inventory gaps, links assets to owners and findings, and prioritizes exposure review, with every discovery action constrained to authorized scope.

Assessment

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Assessment area	Weight	Evidence
Working functionality	**25%**	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	**20%**	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	**20%**	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	**15%**	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	**10%**	Traceable findings, sources, assumptions, logs
Documentation and presentation	**10%**	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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